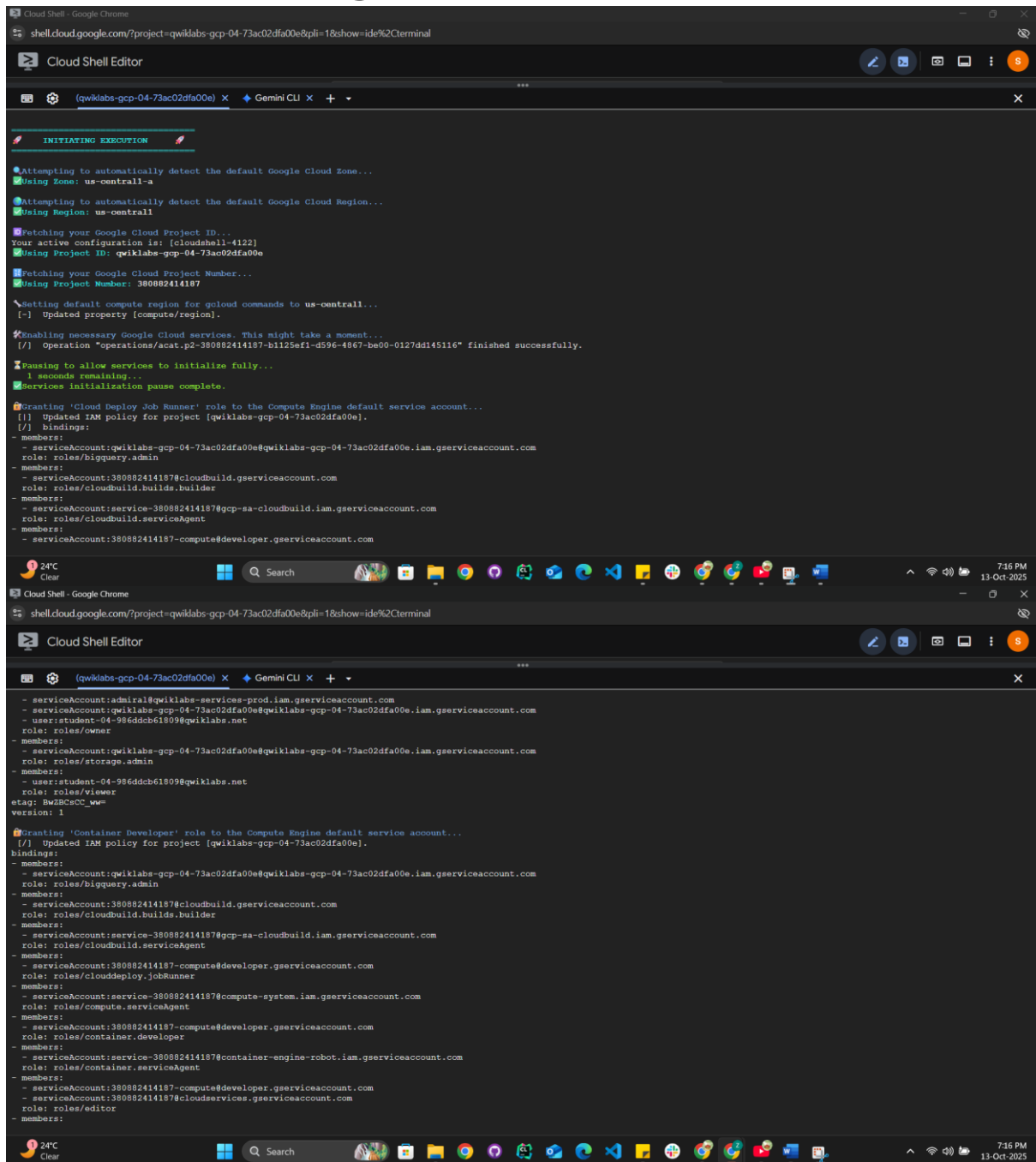


4. Implement CI CD Pipelines on Google Cloud Challenge Lab



```
Cloud Shell - Google Chrome
shell.cloud.google.com/project-qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

INITIATING EXECUTION

Attempting to automatically detect the default Google Cloud Zone...
Using Zone: us-central1-a
Attempting to automatically detect the default Google Cloud Region...
Using Region: us-central1
Fetching your Google Cloud Project ID...
Your active configuration is: [cloudshell-4122]
Using Project ID: qwiklabs-gcp-04-73ac02dfa00e
Fetching your Google Cloud Project Number...
Using Project Number: 380882414187
Setting default compute region for gcloud commands to us-central1...
[-] Updated property [compute/region].
Enabling necessary Google Cloud services. This might take a moment...
[!] Operation "operations/acet.p2-380882414187-b1125ef1-d596-4667-be00-0127dd145116" finished successfully.
Pausing to allow services to initialize fully...
1 seconds remaining...
Services initialization pause complete.
Granting 'Cloud Deploy Job Runner' role to the Compute Engine default service account...
[!] Updated IAM policy for project [qwiklabs-gcp-04-73ac02dfa00e].
bindings:
- members:
  - serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:380882414187@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-380882414187@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com

24°C Clear
Cloud Shell - Google Chrome
shell.cloud.google.com/project-qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

- serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
- serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
- user:student-04-986ddcb61809@qwiklabs.net
  role: roles/owner
- members:
  - serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
  role: roles/storage.admin
- members:
  - user:student-04-986ddcb61809@qwiklabs.net
  role: roles/viewer
etag: Bw2BcCC_wm
version: 1
Granting 'Container Developer' role to the Compute Engine default service account...
[!] Updated IAM policy for project [qwiklabs-gcp-04-73ac02dfa00e].
bindings:
- members:
  - serviceAccount:qwiklabs-gcp-04-73ac02dfa00e@qwiklabs-gcp-04-73ac02dfa00e.iam.gserviceaccount.com
  role: roles/bigquery.admin
- members:
  - serviceAccount:380882414187@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-380882414187@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com
  role: roles/clouddeploy.jobRunner
- members:
  - serviceAccount:service-380882414187@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com
  role: roles/container.developer
- members:
  - serviceAccount:service-380882414187@container-engine-robot.iam.gserviceaccount.com
  role: roles/container.serviceAgent
- members:
  - serviceAccount:380882414187-compute@developer.gserviceaccount.com
  - serviceAccount:380882414187@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
```

```
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

Creating Artifact Registry repository 'cicd-challenge' for Docker images in us-central1...
[!] Note: Create request issued for: [cicd-challenge]
Waiting for operation [projects/qwiklabs-gcp-04-73ac02dfa00e/locations/us-central1/operations/6619079e-cde4-498f-b082-80d7aff09b3f] to complete...
[!] .....done.
[-] Created repository [cicd-challenge].

Creating GKE cluster 'cd-staging' in zone us-central1-a (asynchronously)...
[!] Note: Your Pod address range ('--cluster-ip4-cidr') can accommodate at most 1008 node(s).
[!] NAME: cd-staging
TYPE:
LOCATION: us-central1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:

Creating GKE cluster 'cd-production' in zone us-central1-a (asynchronously)...
[!] Note: Your Pod address range ('--cluster-ip4-cidr') can accommodate at most 1008 node(s).
[!] NAME: cd-production
TYPE:
LOCATION: us-central1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:

Navigating to home directory...
Cloning 'cloud-deploy-tutorials' repository from GitHub...
Cloning into 'cloud-deploy-tutorials'...
[!] remote: Enumerating objects: 1620, done.
remote: Counting objects: 100% (151/151), done.
remote: Compressing objects: 100% (85/85), done.
remote: Total 1620 (delta 103), reused 96 (delta 96), pack-reused 1469 (from 1)
Receiving objects: 100% (1620/1620), 588.13 KiB | 2.24 MiB/s, done.
Resolving deltas: 100% (795/795), done.
Changing directory to 'cloud-deploy-tutorials'...
Checking out a specific commit (c3cae80) silently...
Changing directory to 'tutorials/base'...

Generating Scaffold configuration (scaffold.yaml) from template...
Updating Scaffold configuration with Project ID: qwiklabs-gcp-04-73ac02dfa00e...

24°C Clear
Cloud Shell - Google Chrome
shell.cloud.google.com/?project=qwiklabs-gcp-04-73ac02dfa00e&pli=1&show=ide%2Cterminal

Cloud Shell Editor

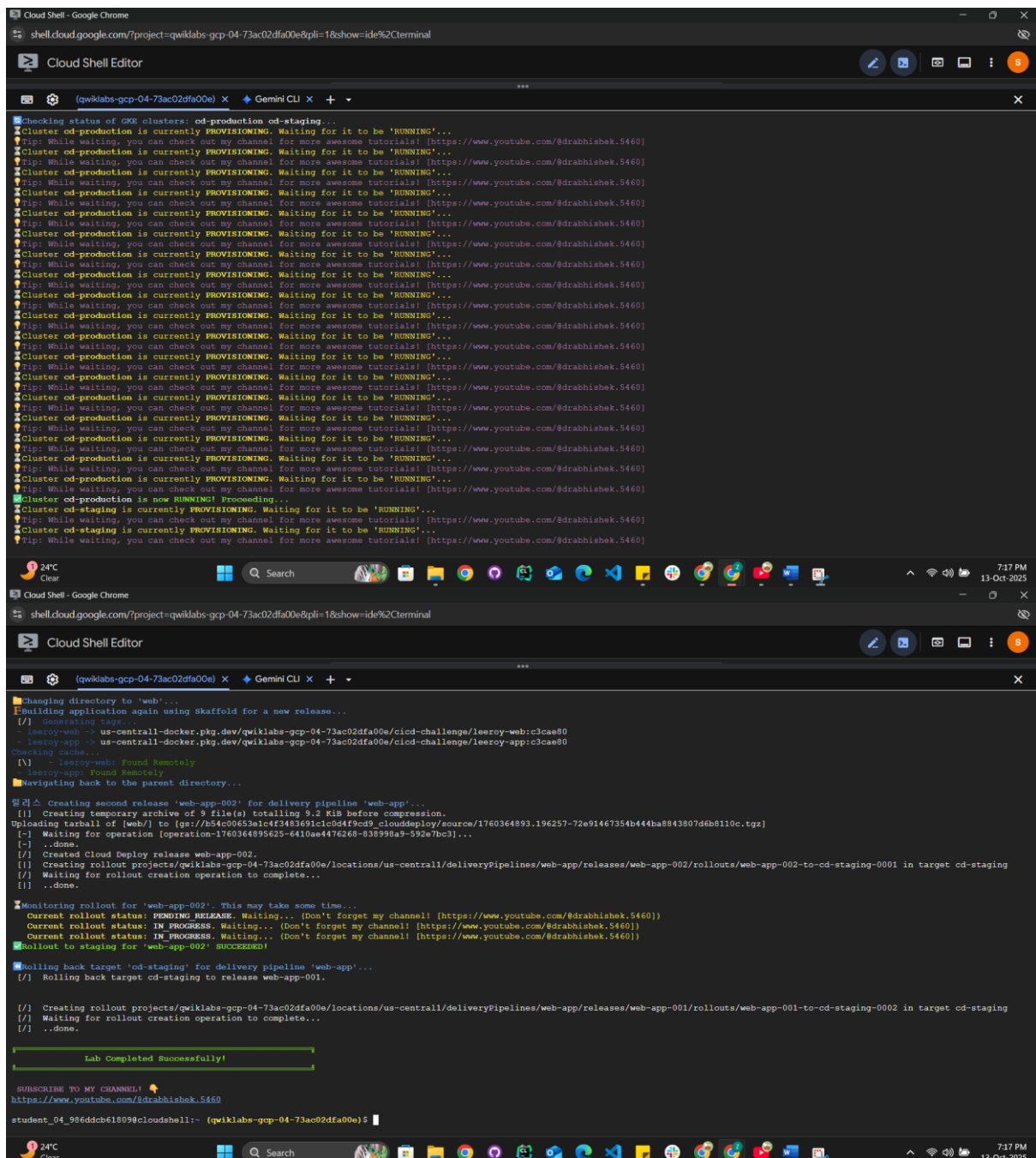
(qwiklabs-gcp-04-73ac02dfa00e) x Gemini CLI x +

Bucket not found. Creating bucket in region us-central1. Open a new tab with project...
[!] Creating gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/...

Changing directory to 'web'...
Building application using Scaffold and outputting artifacts to 'artifacts.json'...
Repository: us-central1-docker.pkg.dev/qwiklabs-gcp-04-73ac02dfa00e/cicd-challenge
[!] Generating logs...
- leeroyw [-] us-central1-docker.pkg.dev/qwiklabs-gcp-04-73ac02dfa00e/cicd-challenge/leeroy-web:c3cae80
- leeroy-app -> us-central1-docker.pkg.dev/qwiklabs-gcp-04-73ac02dfa00e/cicd-challenge/leeroy-app:c3cae80
Checking cache...
[!] - leeroy-web: Not found. Building
- leeroy-app: Not found. Building
Starting build...
Building 2 artifacts in parallel
Building [leeroy-web]...
[!] Pushing code to gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/source/qwiklabs-gcp-04-73ac02dfa00e-e340af6f-858d-4976-a940-105934aa0128.tar.gz
[!] Logs are available at
https://source.cloud.google.com/qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/log-7527769f-0f90-40d0-9884-2a3f6e6ba2d.txt
[!] starting build "7527769f-0f90-40d0-9884-2a3f6e6ba2d"

FETCHSOURCE
Fetching storage object: gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/source/qwiklabs-gcp-04-73ac02dfa00e-e340af6f-858d-4976-a940-105934aa0128.tar.gz#1760364368920099
[!] Copying gs://qwiklabs-gcp-04-73ac02dfa00e_cloudbuild/source/qwiklabs-gcp-04-73ac02dfa00e-e340af6f-858d-4976-a940-105934aa0128.tar.gz#1760364368920099...
/ [1 files] [ 986.0 B / 986.0 B]
Operation completed over 1 objects/986.0 B.

BUILD
Already have image (with digest): gcr.io/cloud-builders/docker
Sending build context to Docker daemon 4.096kB
Step 1/7 : FROM golang:1.12.9-alpine3.10 as builder
[!] 1.12.9-alpine3.10: Pulling from library/golang
9d48c3bd43c5: Pulling fs layer
7f94eaf8af20: Pulling fs layer
9fe9984849c1: Pulling fs layer
cf0db633a67d: Pulling fs layer
0f7136d71739: Pulling fs layer
cf0db633a67d: Waiting
0f7136d71739: Waiting
7f94eaf8af20: Verifying Checksum
7f94eaf8af20: Download complete
9d48c3bd43c5: Verifying Checksum
9d48c3bd43c5: Download complete
9fe9984849c1: Verifying Checksum
```



Software Development & Con...Implement CI/CD Pipelines on...cloudskillsboost.google/paths/76/course_templates/691/labs/594614

Google CloudDashboardExplorePathsSubscriptions

Apply your skills in Google Cloud consoleGet started

10Implement CI/CD Pipelines on Google Cloud

Course · 1 hour 25%15 minutescomplete

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

Implement CI/CD Pipelines on Google Cloud: Challenge Lab

End Lab01:12:28

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Project IDqwiklabs-gcp-04-73ac82d

Usernamestudent-04-986ddcb61889i

Passwordxmv4W9WCd4Xf

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Manual Last Updated May 19 2025

Lab Last Tested May 19 2025

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Check complete. Points earned: 20. Message: Assessment Completed!

Next

Lab instructions and tasks100/100

GSP393

Overview

Setup

Challenge Scenario

Task 1. Prework - Set up environment, enable APIs and create clusters

Task 2. Build the images and upload to the repository

Task 3. Create the Delivery Pipeline

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Course · 1 hour 25%15 minutescomplete

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

Implement CI/CD Pipelines on Google Cloud: Challenge Lab

End Lab01:12:15

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Project IDqwiklabs-gcp-04-73ac82d

Usernamestudent-04-986ddcb61889i

Passwordxmv4W9WCd4Xf

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Manual Last Updated May 19 2025

Lab Last Tested May 19 2025

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Checkpoints

Validate that the container was created and added to the repositoryCheck my progress20 / 20

Verify that the Cloud Deploy targets have been createdCheck my progress20 / 20

Verify the release to the StagingCheck my progress20 / 20

Lab instructions and tasks

Task 1. Prework - Set up environment, enable APIs and create clusters

Task 2. Build the images and upload to the repository

Task 3. Create the Delivery Pipeline

Software Development & Con...Implement CI/CD Pipelines on...cloudskillsboost.google/paths/76/course_templates/691/labs/594614

Google CloudDashboardExplorePathsSubscriptions

Apply your skills in Google Cloud consoleGet started

10Implement CI/CD Pipelines on Google Cloud

Course · 1 hour 15 minutes25% complete

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

Implement CI/CD Pipelines on Google Cloud: Challenge Lab

End Lab01:12:04

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Project ID

qw1k1abs-gcp-04-73ac02d

Username

student-04-986ddcb61809i

Password

xmv4IW9WCd4Xf

Previous

Google Cloud training and certification

...helps you make the most of Google Cloud tech skills and best practices to help you get up to speed on your journey. We offer fundamental to advanced level virtual options to suit your busy schedule. Certification your skill and expertise in Google Cloud technology.

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Verify the release to the Staging environment

Check my progress

20 / 20

Verify the release to the Production environment

Check my progress

20 / 20

Verify the rollback ran successfully

Check my progress

20 / 20

environment, enable APIs and create clusters

Task 2. Build the images and upload to the repository

Task 3. Create the Delivery Pipeline

Next

24°C Clear

Search

7:18 PM13-Oct-2025